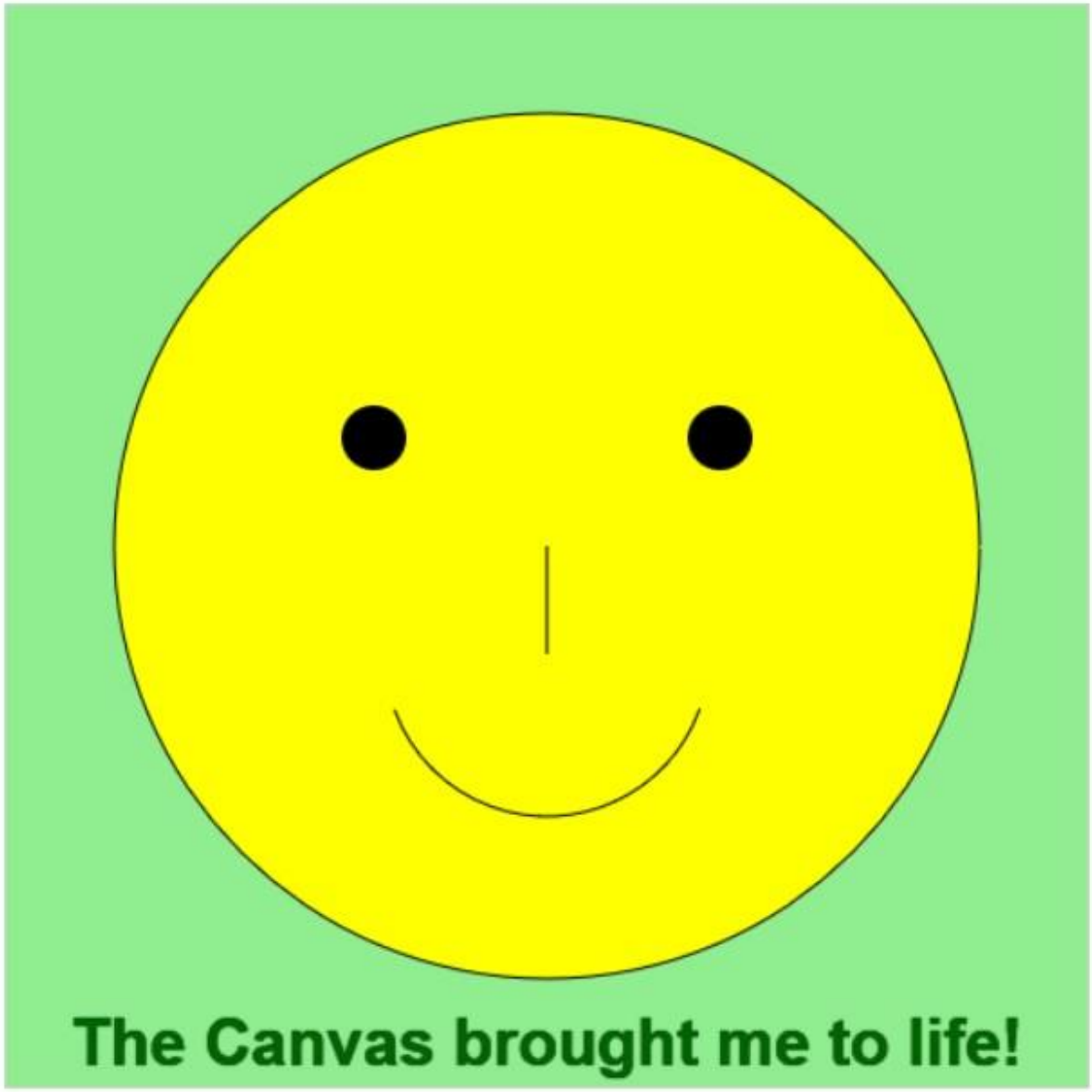
Lab Practical

Exercise 1 – The HTML5 Canvas



Your mission is to recreate the image displayed in the figure. The image features a delightful smiley face and some text, all skillfully drawn using the HTML5 Canvas element and JavaScript. Get ready to unleash your creativity and coding prowess to bring this charming artwork to life!

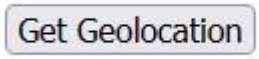
Exercise 2: HTML5 Geolocation

Objective: In this exercise, you will create a webpage that uses the HTML5 Geolocation API to get the client's location and displays this location on the webpage.

Instructions:

Create the HTML structure: Start by creating a new HTML file. In the body of the HTML file, add a button element and a paragraph element. The button should have an onclick attribute that calls a JavaScript function named getGeolocation(). The paragraph should have an id of 'geolocation'. Here's an example of how your HTML structure might look:

```
<button onclick="getGeolocation()">Get Geolocation</button>
<p id="geolocation"></p>
```



Add the JavaScript function: Next, create a new JavaScript file named geolocation.js. In this file, define the getGeolocation() function and showGeolocation () function. The getGeolocation() function should check if the Geolocation API is supported by the client's browser. If it is, it should call the getCurrentPosition() method, which gets the client's current position and passes it to the showGeolocation() function. The showGeolocation() function should update the content of the paragraph element with the id ‘geolocation’ to display the client's latitude and longitude.

Here's an example of how your JavaScript functions might look:

```
let x = document.getElementById("geolocation");

function getGeolocation() {
  if (navigator.geolocation) {
    navigator.geolocation.getCurrentPosition(showGeolocation);
  } else {
    x.innerHTML = "Geolocation is not supported by this browser.";
  }
}

function showGeolocation(position) {
  x.innerHTML = "Latitude: " + position.coords.latitude +
  "<br>Longitude: " + position.coords.longitude;
}
```

getGeolocation(): This function is called to start the process of getting the geolocation.

- **if (navigator.geolocation) {...}:** This line checks if the Geolocation API is supported in the user's browser. The navigator.geolocation object is defined if the API is supported.
- **navigator.geolocation.getCurrentPosition(showGeolocation);**: If the Geolocation API is supported, this line is executed. The getCurrentPosition() method is a part of the Geolocation API and is used to get the current geographical position of the user's device. It takes one argument: a callback function that will be executed when the position data is available. In this case, the callback function is showGeolocation().
- **else { x.innerHTML = "Geolocation is not supported by this browser.";** }: If the Geolocation API is not supported, this line is executed. It updates the HTML of the element with the id 'x' to display a message indicating that geolocation is not supported.

showGeolocation(position): This function is the callback function that is passed to the getCurrentPosition() method. It is called when the position data is available.

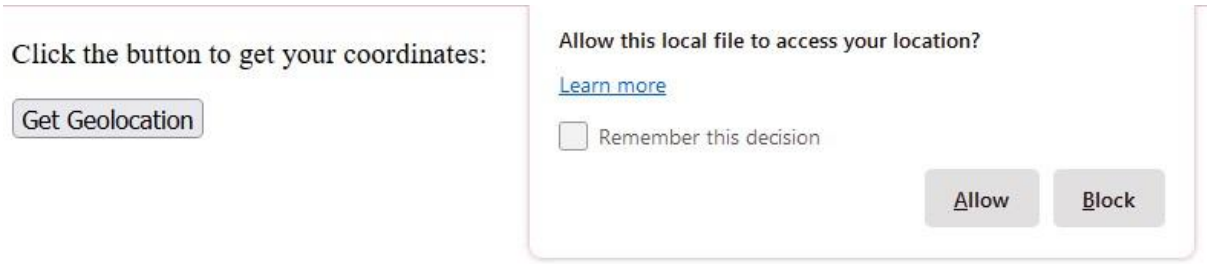
- **position**: This parameter represents the position data that is passed to the function by the getCurrentPosition() method. It is an object that contains various properties related to the position, such as coords.latitude and coords.longitude, which represent the latitude and longitude of the position, respectively.
- **x.innerHTML = "Latitude: " + position.coords.latitude + "
Longitude: " + position.coords.longitude;**: This line updates the HTML of the element with the id 'x' to display the latitude and longitude of the position.

Include the geolocation.js file: Then, in your HTML file, include the geolocation.js file using the <script> tag with the src attribute. The src attribute should be set to the path of the geolocation.js file. If the HTML file and the JavaScript file are in the same directory, the path would just be the name of the JavaScript file. Remember, the path in the src attribute should be the relative path from the HTML file to the JavaScript file. If the JavaScript file is in a different directory, you would need to include the directory in the path. For example, if the JavaScript file is in a directory named 'js', the src attribute would be src="js/geolocation.js".

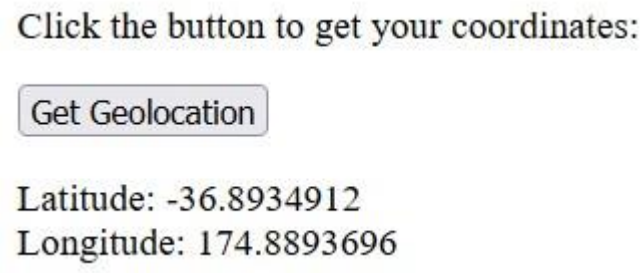
Here's how your HTML file might look:

```
<script src="geolocation.js"></script>
```

Now, when the "Get Geolocation" button is clicked, the getGeolocation() function from the geolocation.js file will be called.



Once the “Allow” is clicked the Latitude and Longitude will be displayed.



Exercise 3: Local Storage Lab: To-Do List with Local Storage

Objective:

Create a simple to-do list where tasks can be added and removed. The list should be saved in Local Storage so that tasks persist even after the browser is closed.

